

Guide: Using an Auxiliary Convergent Orbit

Collatz proof project — September 5, 2026

The proof strategy

For a natural parameter u , the transfer question asks whether convergence of $3u+2$ implies convergence of $27u+20$. Proving this for every parameter would prove Collatz. Strong induction lets us use transfer at any strictly smaller parameter, provided every case is eventually covered.

What the new bridge does

Start with the known-convergent smaller partner. If its orbit reaches $3v+2$ for some $v < u$, the induction hypothesis supplies convergence of $27v+20$. This auxiliary orbit can then meet the original larger partner. The original two partners do not have to return together to a smaller pair.

Lean verifies this rule and the following infinite family:

$$u = 28 + 4096Q, \quad v = 21 + 3072Q < u.$$

The smaller partner reaches $3v+2$ after two steps. The original larger partner after twelve steps and the auxiliary larger partner after ten steps both equal $47 + 6561Q$. These identities hold for every natural Q . Transfer at v therefore implies transfer at u .

What the experiment adds

Among parameters 0 through 1000, 812 pairs enter the trivial cycle without ever meeting the old merge or simultaneous-return criteria. Following those same pairs for longer cannot produce either old certificate. This does not make their transfer statements false: both partners converge. It also does not rule out using finite base cases in an induction.

The expanded search allows up to two auxiliary bridges and thirty ordinary steps. It finds 599 conditional certificates, including 421 of those 812 previously excluded parameters. It leaves 402 parameters at its depth limit. These are finite search results, not a proof that every parameter has a bridge certificate. Larger search limits may change the counts.

What has been checked

Lean checks the general inference rule, the strict decrease in the displayed family, and all its orbit identities for arbitrary Q . The search records every recursive bridge and checks its parameter decreases uniformly over the entire cylinder. Tests replay all saved certificates at four quotients and compare a small complete search with independent path enumeration. The whole census has not been certified in Lean.

What remains open

The smaller transfer premises are still required. We need a proof that all parameters admit a checked base case or a finite certificate whose recursive parameters are smaller. The current search does not establish this coverage. Collatz remains unproved.

Reproducing the work

Build `Collatz.AffineBridge` from `lean/`. From the repository root, run `python3 analysis/affine_bridge_search.py` and `python3 -m unittest discover -s analysis -p test_affine_bridge_search.py`. The accompanying technical paper gives the exact symbolic bridge conditions.